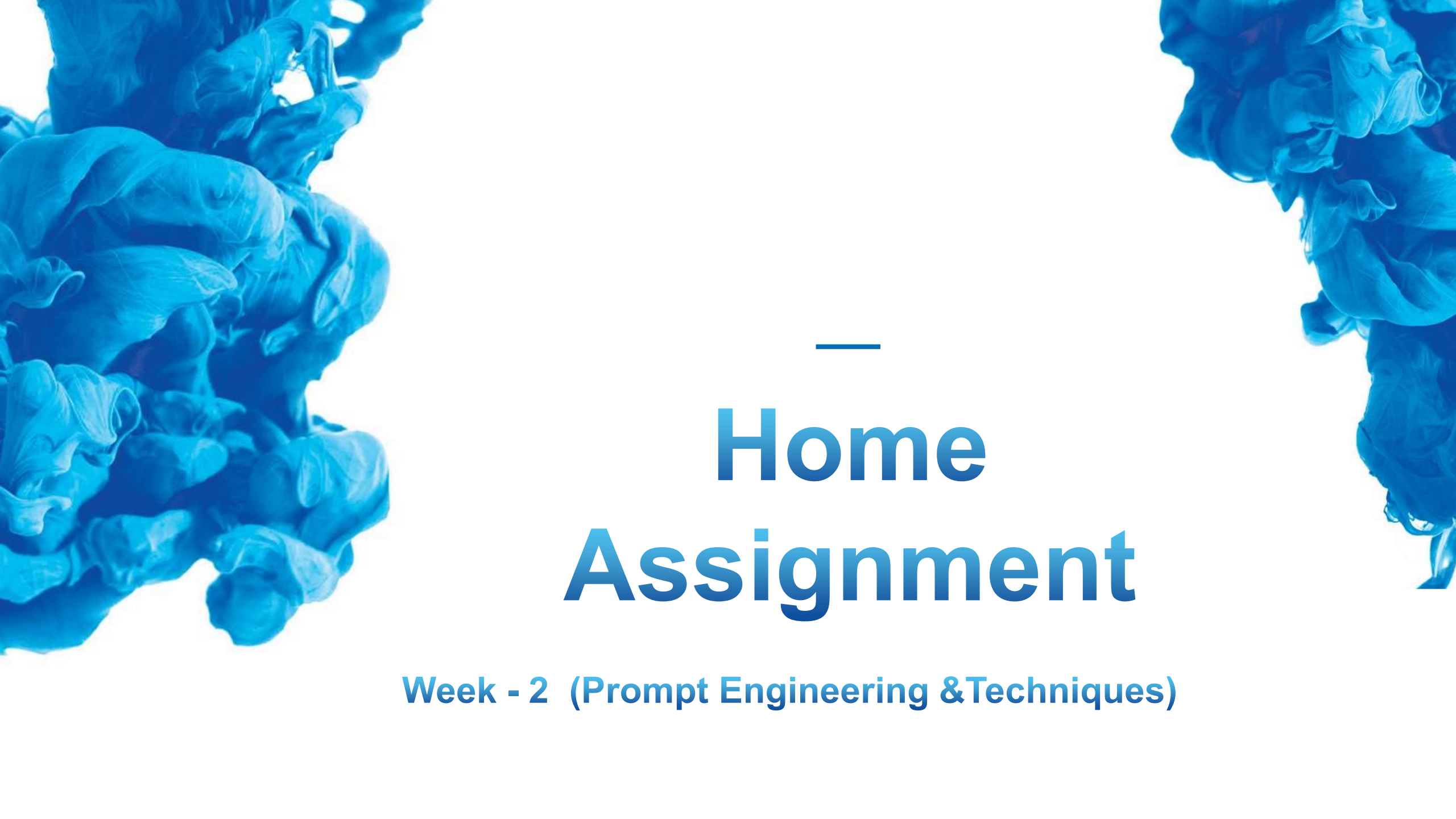


The image features a white background with vibrant blue ink splashes in the top-left and top-right corners. The ink has a fluid, organic quality, with various shades of blue and white creating a sense of movement and depth. In the center of the image, there is a small horizontal blue line above the main title.

— Home Assignment

Week - 2 (Prompt Engineering & Techniques)



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PART 01

ICEPOT TECHNIQUES

Given 3 Scenarios identify what is missing, rebuild the prompt using ICEPOT, run both prompts in an AI tool, and observe the difference in output quality

Scenario 1



Search Functionality

✗ Weak: "Test the search feature."

What's missing: No product type (e-commerce, hospital), no filter/sort behavior, no empty/invalid input scenarios, no performance angle.

Search Functionality

Instructions

Generate comprehensive functional, negative, boundary, and performance test scenarios for the Search feature. Include validation for search behavior, filters, sorting, and error handling.

Persona

Act as a Senior QA Automation Engineer with expertise in web, API, usability, accessibility, and performance testing. Think like a tester preparing regression-ready test scenarios for a production release.

Context

The application is an e-commerce website with product search, category filters, price filters, and sorting options. Consider valid, invalid, empty, special character, partial keyword, and large dataset search scenarios.

Output Format

Present the results in a markdown table with Test Scenario ID, Test Scenario, Expected Result, and Priority. Categorize scenarios into Positive, Negative, Boundary, Performance, Security, and Accessibility.

Examples

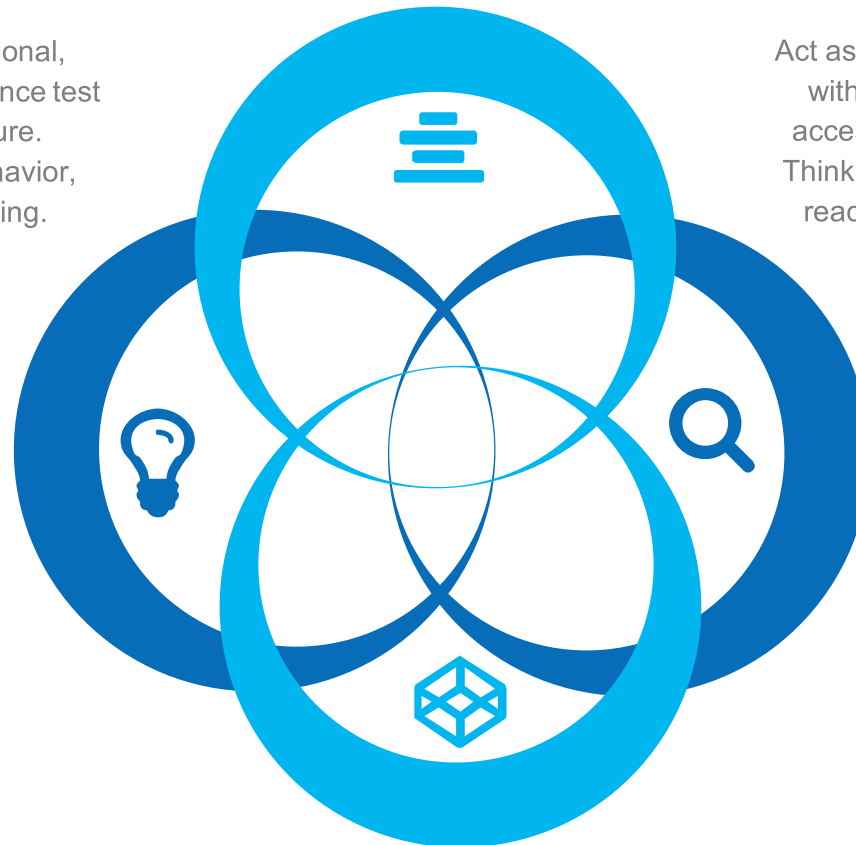
Example: Search "iPhone" and verify relevant products are displayed with correct sorting.

Example: Search an invalid product name and verify a "No products found" message is displayed.

Tone

Keep the response professional, structured, and suitable for JIRA/Xray test case creation.

Ensure the scenarios are concise, realistic, and follow industry QA best practices.



Scenario 2



FILE UPLOAD

✗ Weak: "Generate test cases for file upload."

What's missing: No file types, no size limits, no success/failure criteria, no security angle (malicious file upload).

FILE UPLOAD

Instructions

Generate comprehensive manual and API test cases for the File Upload functionality.

Include positive, negative, boundary, security, usability, and performance scenarios.

Context

The application allows users to upload PDF, JPG, PNG, and DOCX files up to 10 MB. Reject unsupported file types, oversized files, corrupted files, duplicate uploads, and malicious files while displaying appropriate error messages.

Examples

Example valid upload: Resume.pdf (2 MB)
→ Upload succeeds with a success message.

Example invalid upload: malware.exe or image.png (15 MB) → Upload is blocked with the correct validation error.

Persona

Act as a Senior QA Engineer with expertise in web application, API, and security testing.

Create test cases that are suitable for execution and automation.

Output Format

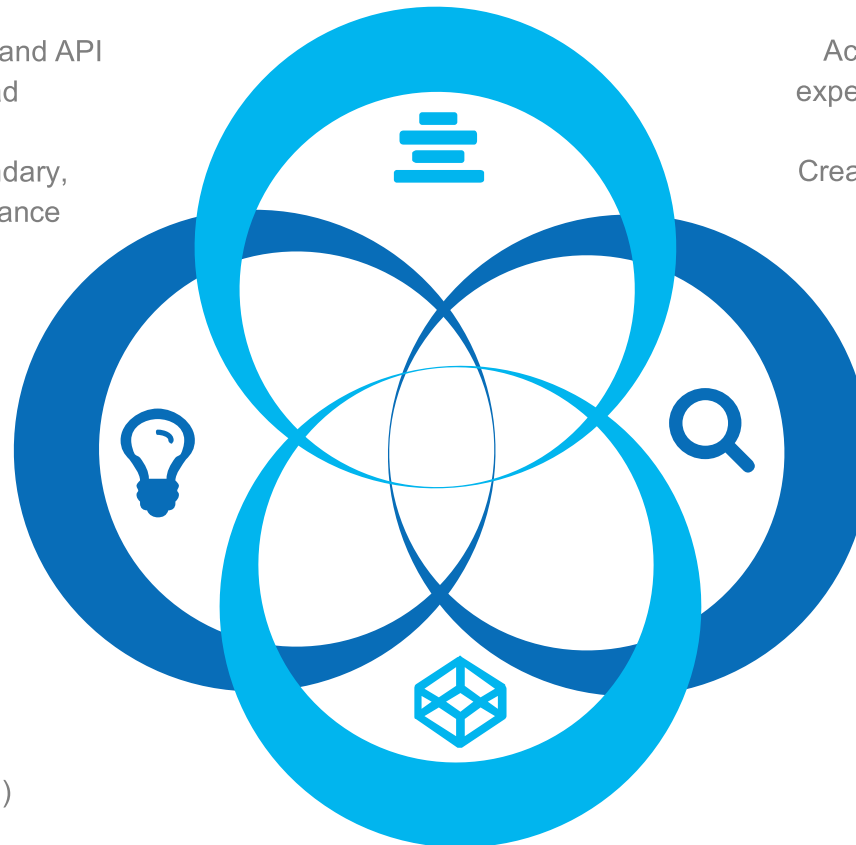
Present the output in a table with Test Case ID, Scenario, Test Steps, Test Data, Expected Result, Priority, and Test Type.

Group the test cases into Functional, Negative, Boundary, Security, and Performance categories.

Tone

Use a professional, structured, and concise testing style.

Ensure the test cases are practical, complete, and aligned with real-world QA standards.



Scenario 3

Role Based Access control

✗ Weak: "Write test cases for user roles."

What's missing: No role types (Admin, Manager, Viewer), no permission matrix, no unauthorized access scenarios, no app domain.

Role Based Access Control

Instructions

Generate comprehensive test cases for a file upload functionality. Include positive, negative, validation, security, and permission-based scenarios.

Persona

Act as a Senior QA Engineer with expertise in API, UI, security, and functional testing for enterprise applications.

Context

The application allows users to upload files. Validate file upload behavior based on supported file types, file size limits, duplicate files, corrupted files, and user access permissions.

Examples

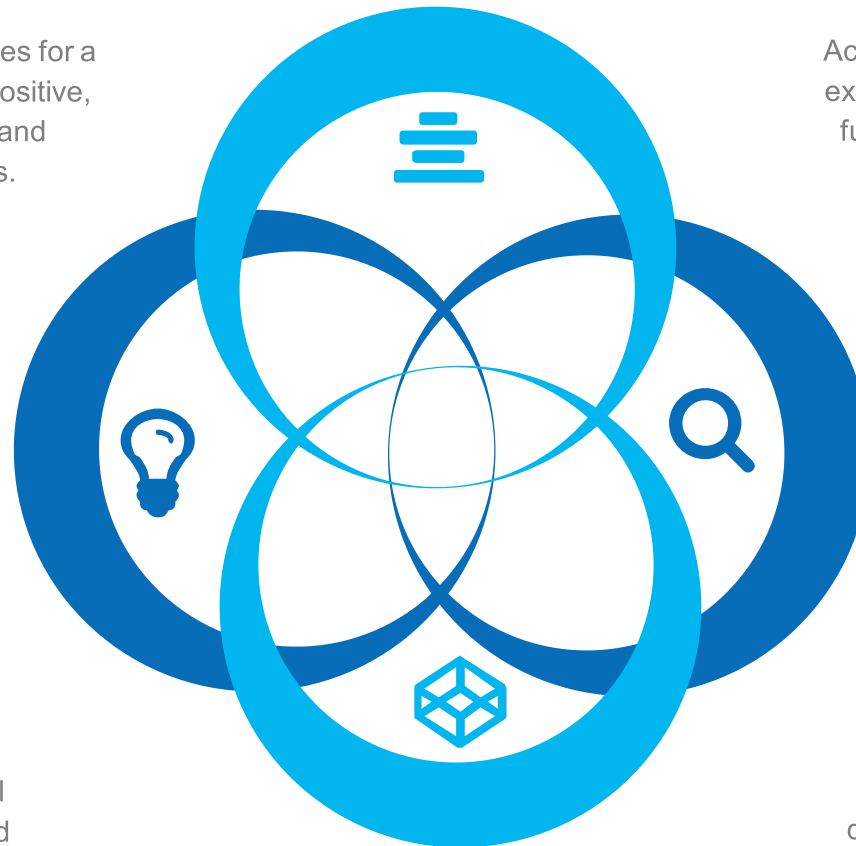
Example scenarios: Admin can upload all allowed files, Viewer has restricted upload access, invalid file formats should be rejected, and oversized files should display validation messages.

Output Format

Provide test cases in a structured format with Test Scenario, Preconditions, Test Steps, Test Data, Expected Result, and Priority.

Tone

Use a professional and detailed QA documentation style suitable for sharing with developers, product owners, and stakeholders.



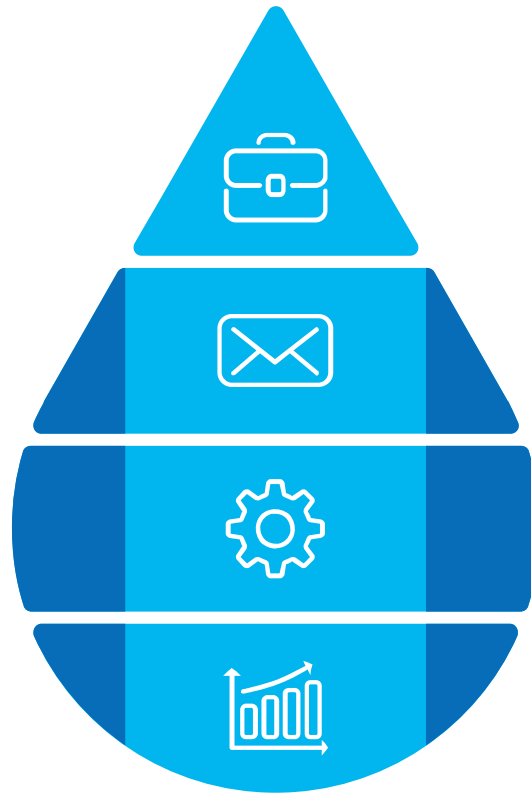
Please enter title content here

Title text addition

The user can demonstrate on a projector or computer, or print the presentation and make it film

Title text addition

The user can demonstrate on a projector or computer, or print the presentation and make it film



Title text addition

The user can demonstrate on a projector or computer, or print the presentation and make it film

Title text addition

The user can demonstrate on a projector or computer, or print the presentation and make it film



PART 02

Socratic Technique

Primarily belongs in the ideation, planning, and problem-diagnosis workflow steps

Socratic Technique



What is

Socratic prompting is the practice of interrogating an AI model output rather than accepting it at face value. Developers use it to surface assumptions, test reasoning, expose edge cases, and improve the quality of generated answers or code.

Why to use

It builds critical thinking: It mimics a great teacher. Instead of handing you the answer, it helps you figure it out yourself. It uncovers hidden assumptions: It challenges your biases. It prevents "AI Hallucinations": It forces the AI to check its own work before giving a final response

When to use

When learning something new: Use this when studying a complex subject (like law, math, or coding) so you truly learn the material instead of just memorizing it.
When you are stuck on a problem: Use this for brainstorming when you are not sure where to start. The AI's questions will help you break down the problem.
When testing a big idea: Use this to critique your own business plan or project.
Ask the AI: "What are the hidden assumptions in my plan?" or "What would make my idea fail?"



"I am struggling with switching my automation testing to AI test engineer. Use the Socratic method to guide me to my own conclusions, but work toward a clear action plan. Ask me about my current skills and situation, and then help me synthesize this into 2-3 actionable next steps.

I'll use a Socratic approach: I'll ask targeted questions first to understand your current position, then we'll convert your answers into a practical transition plan toward an AI Test Engineer role.

Let's start with your current situation:

1. Current Automation Skills

What is your current automation testing experience?

- A) Selenium + Java/TestNG
- B) API automation (REST Assured/Postman)
- C) Mobile automation (Appium/Katalon)
- D) CI/CD and automation framework design
- E) Other skills (please mention)



1. A

Great. You have identified your current automation foundation as:

1. Automation skills: Selenium + Java/TestNG

Let's continue narrowing down your transition path. Your Selenium/TestNG experience is a strong base because AI Test Engineering still needs automation skills, but the focus shifts from **UI behavior validation** to **AI output quality, API validation, evaluation frameworks, and reliability testing**.

Next questions:

2. GenAI exposure:

- A) Only attending GenAI/LLM classes
- B) Understanding concepts like prompts, tokens, embeddings, RAG
- C) Testing LLM APIs using Postman/automation
- D) Building AI test frameworks or evaluation scripts
- E) Working on a real GenAI product

3. Target AI role:

- A) QA Engineer testing AI applications (LLM/API/RAG validation)
- B) Automation Engineer using AI tools for testing





2.A

Good. So far, your profile looks like:

- **Automation foundation:** Selenium + Java/TestNG
- **GenAI exposure:** Currently learning through GenAI/LLM classes

This means your next growth area is likely not basic automation, but learning how AI systems are tested differently from traditional applications.

Let's continue:

3. Target AI role:

- A) QA Engineer testing AI applications (LLM/API/RAG validation)
- B) Automation Engineer using AI tools for testing
- C) AI Quality Engineer building evaluation frameworks
- D) AI/ML Engineer with testing responsibilities

4. Biggest gap:

- A) I understand automation but don't know AI concepts deeply
- B) I understand AI concepts but don't know how to test them
- C) I don't know AI QA tools/frameworks



3.A,C,B

Good insight. Your target direction is actually a combination of three paths:

- **A) QA Engineer testing AI applications** → You want to validate LLM apps, APIs, RAG, prompts, responses.
- **B) Automation Engineer using AI tools** → You want to leverage your Selenium/Java automation background with AI testing.
- **C) AI Quality Engineer building evaluation frameworks** → You are interested in deeper AI quality validation.

This combination is realistic because your existing automation skills can become your advantage. The transition is not about becoming a data scientist; it is about becoming a **QA engineer who understands AI behavior and builds AI quality checks**.

Let's identify your biggest gap now:

4. Biggest gap:

- A) I understand automation but don't know AI concepts deeply
- B) I understand AI concepts but don't know how to test them
- C) I don't know AI QA tools/frameworks
- D) I don't know how to showcase AI testing experience in resume/interviews

5. AI testing experience:

Which areas have you tried so far?

- A) LLM API testing using Postman/API tools
- B) Prompt testing and prompt engineering validation
- C) RAG testing (retrieval accuracy, citations, grounding)
- D) Hallucination and response quality testing



4.C
5.A,B



Great. Based on your answers, here is your current position:

Your AI Test Engineer Transition Profile

✅ Strong foundation

- Selenium + Java/TestNG automation experience
- Traditional QA mindset (functional, regression, automation)

✅ Current AI exposure

- Learning GenAI/LLM concepts
- Started testing:
 - LLM APIs using Postman/API tools
 - Prompt testing and prompt validation

🔍 Main gap identified

- AI QA tools and frameworks

Now let's use the Socratic method to synthesize your next steps.

Reflection 1:

You already know how to automate **expected behavior** in traditional applications.

For AI systems, the challenge changes:

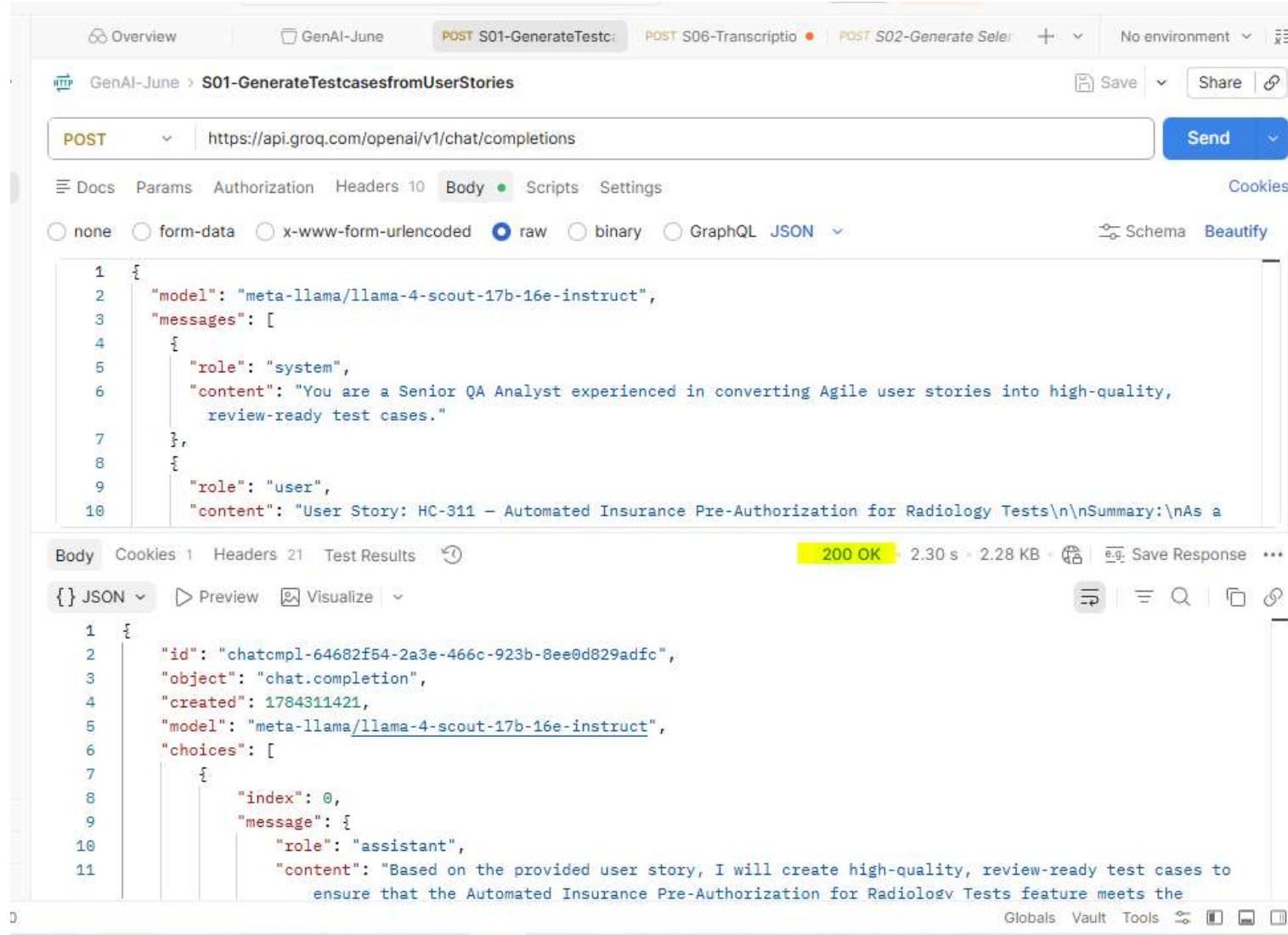
Traditional QA:



"Does the application return the expected value?"

Postman Collection

Header- <https://api.groq.com/openai/v1/chat/completions>



The screenshot displays a Postman interface for a collection named "GenAI-June". The selected item is "S01-GenerateTestcasesfromUserStories", which is a POST request to the endpoint `https://api.groq.com/openai/v1/chat/completions`. The request body is in raw JSON format, containing a system message and a user message. The system message instructs the model to act as a Senior QA Analyst. The user message provides a user story about automated insurance pre-authorization for radiology tests.

The response is a 200 OK status, received in 2.30 seconds with a body size of 2.28 KB. The response body is in JSON format, showing the chat completion details, including the model used and the generated response content.

```
{
  "model": "meta-llama/llama-4-scout-17b-16e-instruct",
  "messages": [
    {
      "role": "system",
      "content": "You are a Senior QA Analyst experienced in converting Agile user stories into high-quality, review-ready test cases."
    },
    {
      "role": "user",
      "content": "User Story: HC-311 - Automated Insurance Pre-Authorization for Radiology Tests\n\nSummary:\nAs a"
    }
  ]
}
```

```
{
  "id": "chatcmpl-64682f54-2a3e-466c-923b-8ee0d829adfc",
  "object": "chat.completion",
  "created": 1784311421,
  "model": "meta-llama/llama-4-scout-17b-16e-instruct",
  "choices": [
    {
      "index": 0,
      "message": {
        "role": "assistant",
        "content": "Based on the provided user story, I will create high-quality, review-ready test cases to ensure that the Automated Insurance Pre-Authorization for Radiology Tests feature meets the"
      }
    }
  ]
}
```

Postman Collection

<https://api.groq.com/openai/v1/audio/transcriptions>

GenAI-June > S06-Transcriptions

POST <https://api.groq.com/openai/v1/audio/transcriptions> Send

Docs Params Authorization Headers 11 Body Scripts Settings Cookies

Pre-request

Post-response

```
3 .tftable {font-size:14px;color:#333333;width:100%;border-width: 1px;border-color: #87ceeb;
4   border-collapse: collapse;}
5 .tftable th {font-size:18px;background-color:#87ceeb;border-width: 1px;padding: 8px;
6   border-style: solid;border-color: #87ceeb;text-align:left;}
7 .tftable tr {background-color:#ffffff;}
8 .tftable td {font-size:14px;border-width: 1px;padding: 8px;border-style: solid;border-color:
9   #87ceeb;}
10 .tftable tr:hover {background-color:#e0ffff;}
11 </style>
12 <table class="tftable" border="1">
13   <tr>
14     <th>ID</th>
15     <td>Object</td>
16   </tr>
17 </table>
```

200 OK 5.11 s • 1.19 KB Save Response

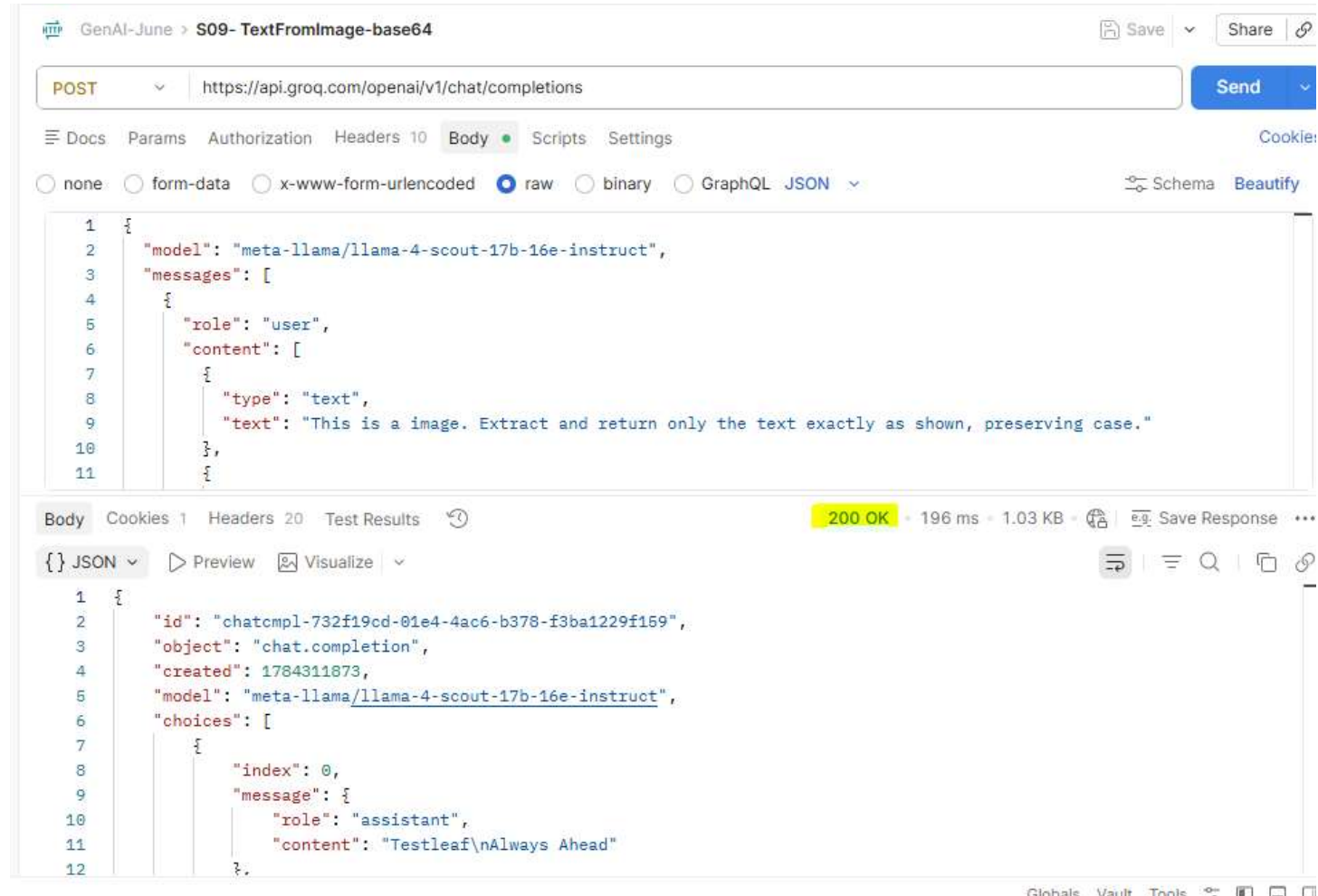
JSON Preview Visualization

```
1 {
2   "text": " Hari here, I'm going to run the daily stand up meeting today. We can get started. Morning Hari, morning
  Sindhu. Radhakrishnan here. So yesterday I've been assigned with a couple of tickets related to balance
  transfer. So I have written manual test cases initially for that and got it signed off from the BA and PO in
  our team. and since the latest build is now available in QA, I have started testing those scenarios and
  monitoring how the system behaves for the different scenarios. So far, the system is good and there are no any
  defects and if there are any, I will raise it accordingly and once the execution is done, I will focus on
  converting those steps into automation scripts and see how it goes. So that's the update from my side. Thanks.
  Thank you Radhakrishnan. Good morning everyone. Sindhu here. Yesterday I have completed smoke and regression
  testing for the payments module and log3 defects in Jira. And today I will be continuing manual testing for
  the reports module and automating the transaction history flow in Selenium. and there is no blockers so far
  and waiting for a new build to proceed with revalidation. So that's all for mv update. Thank you. Okav. Thank
```

Globals Vault Tools

Postman Collection

<https://api.groq.com/openai/v1/chat/completions>





—
Thank you
—

By- JyothiSushma E